BankPulse

PROVIDER COMPARISON · EXISTING WORKFLOW

Chatbot benchmark

Local → OpenAI · Speed, answer quality and cost

Both benchmarks complete 24 Oct 2026 · 159 warm turns per model · 53 fixed turns × 3

Same database snapshot · 45,927 sentences · data through 20 Oct 2026 · base commit `15d1ed8` + uncommitted provider changes

Measure	Target / scale	Local baseline	OpenAI
Answer model	Recorded	Llama 3.1 8B · Q4_K_M	gpt-6-luna
Simple lookups	<5 s	0/51 within target median 12.47 s · correct & on time 0/51	20/51 within target median 5.58 s · correct & on time 20/51
Evidence explanations	≤10 s	0/15 within target median 23.32 s · correct & on time 0/15	15/15 within target median 5.54 s · correct & on time 10/15
Single-topic bank comparisons	≤20 s	14/18 within target median 15.76 s · correct & on time 7/18	18/18 within target median 3.83 s · correct & on time 18/18
Broader, multiple-lookup comparisons	≤20 s	2/9 within target median 26.24 s · correct & on time 0/9	9/9 within target median 9.76 s · correct & on time 3/9
Typical full response · median	Report	12.57 s	4.07 s
Slow full response · 95th percentile	Report	32.69 s	9.46 s
Faithfulness	1–3	2.52/3 126 rated	2.99/3 126 rated
Relevance	1–3	2.57/3 159 rated	2.94/3 159 rated
Correctness	1–3	2.48/3 159 rated	2.93/3 159 rated
Correct and relevant answers	Pass / applicable	99/159 (62%)	148/159 (93%)
Relevant, supported evidence	Pass / applicable	6/24 (25%)	13/24 (54%)
Follow-up turns handled correctly	Pass / applicable	20/33 (61%)	30/33 (91%)
Scope, advice and no-data handling	Pass / applicable	44/45 (98%)	45/45 (100%)
HTTP errors / timeouts	Count	0	0
API cost · full benchmark	US$	$0 hardware/electricity excluded	≈$0.0383 includes probe · excludes pilot

Quality review: Codex-assisted review against actual tool results and the database; **not independent human ratings**. 1 = major issue; 2 = partial / minor issue; 3 = satisfactory. Faithfulness = supported claims; relevance = matches request; correctness = right facts and periods. Prepared guard replies have faithfulness N/A.

What changed

- **Faster:** median response 12.57 s → 4.07 s (3.1× faster). Simple lookups meet 5 s in 20/51 turns.
- **More complete:** answer pass rate 99/159 (62%) → 148/159 (93%); overall queries and follow-ups are more consistent.
- **Still needs work:** 11 OpenAI turns miss the quality bar, mainly incomplete or unhelpful evidence. A provider switch alone does not resolve retrieval quality.

Comparison limit: local sees full local evidence; OpenAI withholds Reddit, CNA, broker-research and unknown-source text, links/citations, and earlier assistant answers. Evidence availability differs, so this measures the configured systems rather than isolating model ability. OpenAI timings include internet latency.

Complete FastAPI response time includes local intent classification, tools, generation and verification; browser rendering is excluded. First-request probes: local 25.86 s, OpenAI 6.87 s, excluded above. OpenAI had a prior pilot, so its probe is not a cold-start comparison. Cost is a token-based estimate, not an invoice.